

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO3

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Supervised Machine Learning

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the impurity measure a decision tree may use in place of entropy.

- A) Gini impurity
- B) Mean squared error
- C) Cosine similarity
- D) Euclidean distance

Q2. Identify the term for a model that simply stores the training data and does its work at prediction time.

- A) An eager learner
- B) A lazy learner
- C) An ensemble learner
- D) A generative learner

Q3. State the shape of the decision boundary that logistic regression produces in the feature space.

- A) A closed circle around each class
- B) A curve of arbitrary shape
- C) A straight line or flat plane
- D) A set of axis-parallel rectangles

Q4. Name the sampling scheme used to build each individual tree of a Random Forest.

- A) Stratified sampling without replacement
- B) Systematic sampling of every fifth row

- C) Cluster sampling of the training set
- D) Bootstrap sampling with replacement

Q5. Identify the value a regression tree predicts at one of its leaves.

- A) The largest value in that leaf
- B) The mean of the training records in that leaf
- C) The number of records in that leaf
- D) The depth of that leaf in the tree

Q6. State what the intercept of a fitted linear regression represents.

- A) The change in the target per unit of a feature
- B) The proportion of variance the model explains
- C) The average error the model makes
- D) The predicted value when every feature is zero

Q7. Explain why a decision tree is described as an interpretable model.

- A) The path from root to leaf reads as a chain of conditions
- B) It always achieves a higher accuracy than others
- C) It requires no training data to be supplied
- D) It stores the whole training set for inspection

Q8. Distinguish underfitting from overfitting by the pattern of the two error readings.

- A) Overfitting has both errors high, underfitting has training low
- B) Both of the two show a high training error
- C) Underfitting has both errors high, overfitting has training low and test high
- D) Both of the two show a low test error

Q9. Describe what information gain measures at a candidate split.

- A) The number of records the split separates
- B) The depth the tree will reach after the split
- C) The reduction in impurity the split achieves
- D) The accuracy the finished tree will attain

Q10. Interpret a fitted linear regression coefficient of - 2.5 on a feature.

- A) The target falls by 2.5 for each unit that feature rises
- B) The target rises by 2.5 for each unit that feature rises
- C) The feature explains 2.5 per cent of the variance
- D) The model makes an average error of 2.5 units

Q11. Predict the effect on variance of averaging the predictions of many different trees.

- A) The variance rises while the bias falls sharply
- B) Both the variance and the bias fall to zero
- C) Neither the variance nor the bias is affected
- D) The variance falls while the bias stays roughly the same

Q12. Recognize the model described: it predicts the majority class among the K training records closest to the query.

- A) Logistic regression
- B) K-Nearest Neighbour
- C) Naive Bayes
- D) Linear regression

Q13. Explain why Naive Bayes performs well on text classification despite its assumption.

- A) Words in a document really are independent
- B) Text data contains no correlated features
- C) It requires an enormous amount of training text
- D) The highest-scoring class is often right even when the probabilities are not

Q14. Summarize what the bias component of a model's error describes.

- A) Error from the model changing with the sample
- B) Error from the noise present in the measurements

- C) Error from assumptions too simple for the true pattern
- D) Error from the features being on different scales

Q15. Distinguish a regression tree from a classification tree.

- A) One predicts a class at its leaves, the other a number
- B) One predicts a number at its leaves, the other a class
- C) Both of the two predict a class at their leaves
- D) Both of the two predict a number at their leaves

Q16. Explain why a polynomial regression of very high degree fits the noise of the training data.

- A) It has enough flexibility to pass close to every point
- B) It reduces the number of features considered
- C) It forces the relationship to be a straight line
- D) It removes the extreme values from the data

Q17. Predict what happens to a K-Nearest Neighbour classifier when many irrelevant features are added.

- A) The distances become sharper and accuracy rises
- B) The distances lose contrast and the neighbours become arbitrary
- C) The classifier automatically ignores those features
- D) The classifier can no longer be trained at all

Q18. Describe what Laplace smoothing prevents in a Naive Bayes classifier.

- A) A single unseen word driving the whole score to zero
- B) The classifier from being trained on text data
- C) The features from being treated as independent
- D) The prior probabilities from being computed

Q19. Explain why a Random Forest does not require its features to be placed on a common scale.

- A) It converts every feature to the same units
- B) It computes distances between the records
- C) It uses only the largest-ranging feature
- D) Each split compares values within a single feature

Q20. Distinguish the parameters of a model from its hyperparameters.

- A) Hyperparameters are learned, parameters are set beforehand
- B) Both of the two are learned from the training data
- C) Parameters are learned from the data, hyperparameters are set beforehand
- D) Both of the two are fixed before training begins

Q21. Apply the fitted model $y = 3 + 1.5x$ to an input of $x = 8$ and determine the predicted value.

- A) 15
- B) 12
- C) 13.5
- D) 16.5

Q22. Solve the following: a regression line has a slope of 2 and passes through the point with coordinates 5 and 16. Determine its intercept.

- A) 10
- B) 26
- C) 6
- D) 3.2

Q23. Use a logistic regression model whose linear score for an observation is 0.8, and determine the class predicted at a threshold of 0.5.

- A) The negative class
- B) The positive class
- C) Neither class
- D) Both classes equally

Q24. Apply a 5-Nearest Neighbour classifier to a query whose five closest training points carry the labels A, A, B, A and B, and determine the prediction.

- A) B
- B) Either A or B at random
- C) No prediction is possible
- D) A

Q25. Examine a regression fit whose residuals are scattered randomly about zero with no pattern, and determine what it indicates.

- A) The model has clearly overfitted the data
- B) A higher-degree polynomial is required
- C) The linear form is appropriate for this data
- D) The features must be rescaled before fitting

Q26. Compare a decision tree limited to a depth of one with an unlimited tree on bias and on variance.

- A) The shallow tree has high variance, the unlimited tree high bias
- B) Both of the two have high bias and low variance
- C) Both of the two have low bias and low variance
- D) The shallow tree has high bias, the unlimited tree high variance

Q27. Solve the following: at a candidate split a node holding 60 samples divides into children of 45 and 15. Determine the fraction that goes to the larger child.

- A) 0.75
- B) 0.25
- C) 0.45
- D) 0.60

Q28. Apply Naive Bayes to a two-class problem in which the unnormalised score for class A is 0.006 and for class B is 0.024, and determine the prediction.

- A) Class A
- B) Class B
- C) Neither class
- D) Both are equally likely

Q29. Use a learning curve on which the training error is very low while the validation error remains high, and determine the diagnosis.

- A) The model is underfitting the training data
- B) The model has an ideal level of complexity
- C) The training set is far too large for the model
- D) The model is overfitting the training data

Q30. Compute the R squared of a regression whose residual sum of squares is 20 and whose total sum of squares is 100.

- A) 0.2
- B) 0.8
- C) 5.0
- D) 0.02

Q31. Apply a Random Forest of 100 trees in which 60 vote Faulty and 40 vote Healthy, and determine the prediction.

- A) Healthy
- B) Neither of the two
- C) Faulty
- D) The vote is a tie

Q32. Examine a K-Nearest Neighbour classifier with K set to 1 on noisy data and determine its behaviour.

- A) A jagged boundary that follows the noise
- B) A smooth boundary that ignores the noise
- C) A straight boundary through the feature space
- D) A prediction of the majority class everywhere

Q33. Apply polynomial regression of degree one and determine what model it is equivalent to.

- A) Ordinary linear regression
- B) A decision tree regressor
- C) A K-Nearest Neighbour regressor
- D) A logistic regression classifier

Q34. Use the sigmoid function to determine the predicted probability when the linear score is exactly zero.

- A) 0
- B) 1
- C) It is undefined
- D) 0.5

Q35. Compare the Gini impurity of a node holding one class alone with that of a node split evenly between two classes.

- A) 0.5 for the pure node and zero for the even one
- B) Zero for the pure node and 0.5 for the even one
- C) Zero for both of the two nodes
- D) One for both of the two nodes

Q36. Apply bootstrap sampling to a training set of 100 records and determine the size of each sample drawn.

- A) 50 records, drawn without replacement
- B) 100 records, drawn without replacement
- C) 100 records, drawn with replacement
- D) 10 records, drawn with replacement

Q37. Examine a model recording 95 per cent accuracy on the training set and 94 per cent on the test set, and determine what it indicates.

- A) The model has badly overfitted the training set

- B) The model has badly underfitted the training set
- C) The model generalises well to unseen data
- D) The test set must have leaked into training

Q38. Use the bias-variance idea to determine the effect of steadily increasing the complexity of a model.

- A) Bias rises while variance falls
- B) Bias falls while variance rises
- C) Both bias and variance fall together
- D) Both bias and variance rise together

Q39. Apply a decision threshold of 0.5 to the predicted probabilities 0.72, 0.31 and 0.55, and determine how many records are called positive.

- A) 1
- B) 3
- C) 0
- D) 2

Q40. Relate the number of trees in a Random Forest to the stability of the predictions it makes.

- A) More trees give steadier predictions with diminishing returns
- B) More trees make the predictions less steady
- C) The number of trees has no effect on stability
- D) More trees always raise the accuracy without limit

----- End of Sample Question Set -----